

Erdős Problem #201

EP-201	L1 - Tractable	Ouvert	Number Theory
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Énoncé original (anglais, reproduction mot pour mot)

Let $G_k(N)$ be such that any set of N integers contains a subset of size at least $G_k(N)$ which does not contain a k -term arithmetic progression. Determine the size of $G_k(N)$. How does it relate to $R_k(N)$, the size of the largest subset of $\{1, \dots, N\}$ without a k -term arithmetic progression? Is it true that $\lim_{N \rightarrow \infty} \frac{R_3(N)}{G_3(N)} = 1$?

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